

Bottom-up build (green)
EUBUCCO footprints, TABULA heat intensities, NUTS3 useful heat

Top-down benchmark (blue)
Hotmaps and census building-type shares, reconciliation check

Useful-heat demand to 2050
LMDI multi-driver trajectory: power growth, household-size decline, envelope renovation
Eq. demand

Technology, cost and policy layers
eight technologies, levelised cost and LCOH, ETS2 carbon, bans and mandates

Four scenarios, Monte Carlo (N = 200 per scenario)
Current Policies / Stated Policies / Net Zero / H2 Push
tab. scenarios

Peak dispatch, peaking investment and network capital

Winter-peak merit order
cold-snap slice dispatched among gas, hydrogen and heat-pump peakers, at an endogenous scarcity power price
Eq. peakers

Peaking-investment test
inframarginal rent over the slice hours versus annualised peaker capital (the missing-money condition)
Eq. missingmoney

Network capital bill
grid reinforcement, district-heat expansion and hydrogen network, by pathway
Eq. infrabill

Explicit 8760-hour dispatch
temperature-derated heat-pump load and Dunkelflaute, power coupling; robustness of the peak findings
Eq. dispatch

Outputs
technology mix and emissions, LCOH, hydrogen peak slice it can win, missing-money gap and infrastructure bill